How Often Do Powers of 2 Start with a 1?

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1 Introduction

Suppose one were to start listing out powers of two. $2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16, 2^5 = 32, 2^6 = 64, 2^7 = 128, 2^8 = 256, 2^9 = 512, 2^{10} = 1024, ...$

After listing the 10 first cases, we notice that 3 of them begin with a 1. What can be said of this pattern? How often do we encounter powers of 2 that start with a 1? Or we could rephrase this problem: given any arbitrary power of 2, what is the probability that this number starts with 1?

Let's consider the first n powers of 2: 2^1, 2^2, ..., 2^n. Suppose among these n powers, there are a_n that begin with a 1 when written down. That would mean that the probability that an arbitrarily chosen number from 2^1, 2^2, ..., 2^n starts with a 1 is $\frac{a_n}{n}$. This probability is dependent upon n, or the number of terms that we are considering. Thus, the limit

$$p_1 = \lim_{n \to \infty} \frac{a_n}{n}$$

(if it exists) will be the probability that an arbitrary power of 2 starts with a 1.

Now we turn to calculating this limit. Notice that among the two-digit numbers, the only power of 2 that starts with 1 is 16, and among the three-digit numbers, the only power of 2 that stats with 1 is 128. The same turns out to be true for four-digit numbers as well: 1024 is the only power of 2 that starts with 1. A question arises: does this pattern hold for all k-digit numbers?

Consider the smallest k-digit power of 2. This number must start with a 1. We can show this is true by contradiction.

Supposing the smallest k-digit power of 2 didn't start with a 1, we would be able to divide that number by 2, reaching another k-digit power of 2 that has a smaller leading digit. We can descend this way, and we will always reach a power of 2 that starts with 1. And we can also see that another k-digit power of 2 with a leading digit of 1 will not exist: the next power of 2 will have a leading digit of 2 or 3, and will continue likewise until we reach the $(k+1)$-digit power of 2 that has a leading digit of 1.

From here, it follows that if 2^n is an m-digit number, then among the numbers $2^1, 2^2, ..., 2^n$ there will be exactly $m-1$ that have a leading digit of 1. This implies that $a_n = m - 1$.
But the claim that 2^n is an is an m-digit number can be rewritten as

$$10^{m-1} \leq 2^n < 10^m \Rightarrow m - 1 \leq n \log 2 < m.$$

Recalling that $a_n = m - 1$, we obtain that $a_n \leq n \log 2 < a_n + 1$. From the left inequality, we get $\frac{a_n}{n} \leq \log 2$, and from the right inequality, we get that $\frac{a_n}{n} > \log 2 - \frac{1}{n}$. Combining this information, we see that $\log 2 - \frac{1}{n} < \frac{a_n}{n} \leq \log 2$.

Utilizing the Squeeze Theorem, we conclude that

$$p_1 = \lim_{n \to \infty} \frac{a_n}{n} = \log 2 \approx 0.30103...$$

This remarkable result shows that about 30% of all powers of 2 have a leading digit of 1.

2 Exercises

1. Let p_i represent the probability that an arbitrarily chosen power of 2 has a leading digit of i. Prove the following: $p_2 + p_3 = p_1$, $p_4 + p_5 = p_2$, $p_6 + p_7 = p_3$, and $p_8 + p_9 = p_4$.

2. Show that $p_4 = 1 - 3 \log 2 \approx 0.096...$

3 The Generalization

A natural question arises after this problem: what is the probability that a power of an arbitrary natural number k has a leading digit of q?

Suppose the number k^n has a leading digit of q: $q \cdot 10^m \leq k^n < (q + 1) \cdot 10^m$, where m is a natural number. Dividing this inequality by $q \cdot 10^m$ and taking the base-10 logarithm of both sides, we get that $0 \leq n \log k - \log q - m < \log \frac{q+1}{q}$. Because the base-10 logarithm is an increasing function, we have that for the upper bound, $\log \frac{q+1}{q} = \log 1 + \frac{1}{q} \leq \log 2 < \log 10 = 1$. Thus, by the inequality above, $n \log k - \log q - m$ is bounded between 0 and 1. And since m is a natural number, it follows that the fractional part of $n \log k - \log q$ is less than $\log \frac{q+1}{q}$.

A reminder that the fractional part of a number x is denoted $\{x\}$, and is defined as $x - \lfloor x \rfloor$. Conversely, if the fractional part of $n \log k - \log q$ is less than $\log \frac{q+1}{q}$, then k^n has a leading digit of n.

By showing this biconditional, we have illustrated the original problem is equivalent to the following: What is the probability that for an arbitrarily chosen number n, the inequality $\{n \log k - \log q\} < \log \frac{q+1}{q}$ holds. To solve this problem, we prove the following theorem.

3.1 Theorem on Fractional Parts

Let α, β be real numbers, where α is irrational, and I is some region length h contained within the interval [0, 1]. Consider the sequence $\alpha + \beta, 2\alpha + \beta, ..., n\alpha + \beta, ...$ Then the probability that an arbitrary term of this sequence has a fractional part that belongs to to I is h.

3.2 Proof

Let r be a natural number. The points $\frac{1}{r}, \frac{2}{r}, ..., \frac{r-1}{r}$ divide the interval [0, 1] into r segments, each of length $\frac{1}{r}$. We will now apply the combinatorial Pigeonhole Principle on these r subsections. Consider the numbers $\{\alpha\}$, $\{2\alpha\}$, ... , $\{(r + 1)\alpha\}$. They all fall into the interval [0, 1]. But since there are $r + 1$ pigeons are r holes in this scenario, then there will at least exist two pigeons that fall into the same hole. That is, there will exist some p' and $p''(p' > p''$ without a loss of generality) such that the fractional parts of $p'\alpha$ and $p''\alpha$ differ by less than $\frac{1}{r}$. Thus, $|p'\alpha - p''\alpha - m| < \frac{1}{r}$ for some whole number m. Substituting $p = p' - p''$, we can say that $|p\alpha - m| = \varepsilon$, where $\varepsilon < \frac{1}{r}$. By the irrationality of α, we note that $\varepsilon \neq 0$.

Now, let s be the smallest natural number that exceeds $\frac{1}{\varepsilon}$, or $s - 1 \leq \frac{1}{\varepsilon} < s$. Examine s consecutive terms of the arithmetic sequence with a common difference of $p\alpha$: $\gamma + p\alpha, \gamma + 2p\alpha, ..., \gamma + sp\alpha$, where γ is some real number. Denote the fractional parts of these numbers as $x_1, x_2, ..., x_s$. By the equality $|p\alpha - m| = \varepsilon$, the numbers $x_1, x_2, ..., x_s$ differ by ε. And by the inequality $\frac{1}{s} < \epsilon \leq \frac{1}{s-1}$, we can conclude that the distance between x_1 and x_s is less than ε.

It follows from here that that number of points that fall into the region I of length h is between the values $\frac{h}{\varepsilon} - 1$ and $\frac{h}{\varepsilon} + 2$, that is, between $sh - 2$ and $sh + 2$ (noting that $h < 1$).

Now take ps consecutive terms of an arithmetic sequence $b_1, b_2, ...b_{ps}$ with a common difference of α, and let is write this sequence out as a table.

$$\begin{bmatrix} b_1 & b_{p+1} & b_{2p+1} & \dots & b_{(s-1)p+1} \\ b_2 & b_{p+2} & b_{2p+2} & \dots & b_{(s-1)p+2} \\ b_3 & b_{p+3} & b_{2p+3} & \dots & b_{(s-1)p+3} \\ \dots & \dots & \dots & \dots & \dots \\ b_p & b_{2p} & b_{3p} & \dots & b_{sp} \end{bmatrix}$$

Every row of this table forms a sequence of the form $\gamma + p\alpha$, as described in the previous paragraph. Since the number of rows equals p, then the number of terms from $b_1, b_2, ...b_{ps}$ that fall into the interval I is between $p(sh - 2)$ and $p(sh + 2)$.

Finally, let n be a natural number that exceeds rps. Dividing n by ps with remainder, we get $n = lps + q$, where $l \geq r$, $0 \leq q < ps$.

It follows that from the numbers $\alpha + \beta, 2\alpha + \beta, ..., n\alpha + \beta$, one can take ps consecutive terms of the arithmetic progression with common difference α l times, and q numbers will remain.

Therefore, $lp(sh - 2) \leq a_n \leq lp(sh + 2) + q$, where a_n is the number of terms from $\alpha + \beta, 2\alpha + \beta, ..., n\alpha + \beta$, the fractional parts of which lie in I. This inequality can be written as $(n - q)h - 2lp \leq a_n \leq (n - q)h + 2lp + q$ from where (by the inequalities $0 \leq h < 1$ and $q < ps$), we get $nh - 2lp - ps \leq a_n \leq nh + 2lp + ps$, that is $|\frac{a_n}{n} - h| \leq \frac{2lp+ps}{n} \leq \frac{2lp+ps}{lps} = \frac{2}{s} + \frac{1}{l} < \frac{3}{r}$ (since $l \geq r$ and $\frac{1}{s} < \varepsilon < \frac{1}{r}$. Thus, there exists a natural number N, namely $N = rps$, such that for $n > N$, $|\frac{a_n}{n} - h| < \frac{3}{r}$. Since r was chosen arbitrarily and can be made arbitrarily large, it follows that

$$\lim_{n \to \infty} \frac{a_n}{n} = h$$

which concludes the proof.

3.3 Resolution to the General Problem

We now may finish the general question: what is the probability that an arbitrarily chosen power of the natural number k has a leading digit of q?

Previously, we reduced this problem to the probability that for an arbitrarily chosen number n, the inequality $\{n \log k - \log q\} < \log \frac{q+1}{q}$ holds.

Note that if k itself is any power of 10 (1, 10, 100, ...), then all of its powers have a leading digit of 1, and the problem is trivial. But if k is not a power of 10, then $\alpha = \log k$ is an irrational number (as an exercise, prove this!).

From the Theorem on Fractional Parts, the probability that an arbitrarily chosen number from the sequence $n \log l - \log q (n = 1, 2, 3, ...)$ has a fractional part that belongs to $[0, \log \frac{q+1}{q}]$ equals $\log \frac{q+1}{q}$. And this is the general solution! The probability that an arbitrarily chosen power of the natural number k has a leading digit of q is $\log \frac{q+1}{q}$.

Note that, surprisingly, the answer is independent of the base of exponentiation, k.

4 Additional Applications of the Theorem

4.1 Problem 1

Let α be an irrational number. Prove that there exists a natural number n such that $\cos n\alpha\pi > 0.999$.

We note that the inequality $\cos n\alpha\pi > 0.999$ is equivalent to $2m\pi - \arccos 0.999 < n\alpha\pi < 2m\pi + \arccos 0.999$ for some whole number m. Thus, $m < \frac{\alpha}{2}n + \frac{1}{2\pi} \arccos 0.999 < m + \frac{1}{\pi} \arccos 0.999$. In other words, the inequality $\cos n\alpha\pi > 0.999$ holds true if and only if the fractional part $\{\frac{\alpha}{2}n + \frac{1}{2\pi} \arccos 0.999\}$ belongs to the interval $[0, \frac{1}{\pi} \arccos 0.999]$. By the theorem on fractional parts, an arbitrarily chosen power of n follows this condition with a probability of $\frac{1}{\pi} \arccos 0.999 \approx 0.014$. Therefore, for a sufficiently large N, approximately 1.4% of the numbers $1, 2, ..., N$ satisfy the conditions of the problem.

4.2 Exercise 2

Find the probability that any arbitrary power of the number k starts with the combination of numbers $q_1 q_2 ... q_r$?

4.3 Exercise 3.1

Prove that for an arbitrary power of the number k, the probability that the j-th number from the left is q is equal to

$$\sum_{i=10^j-2}^{10^{j-1}-1} \log(1 + \frac{1}{q + 10i})$$

4.4 Exercise 3.2

Show that

$$\lim_{j\to\infty} \sum_{i=10^j-2}^{10^{j-1}-1} \log(1 + \frac{1}{q + 10i}) = \frac{1}{10}$$

Hint: First show that $\ln(1+x) < x$ for $x > 0$.

Then use the following: $\log(1+\frac{1}{10i)} - \log(1+\frac{1}{10i+q}) < \log(1+\frac{q}{100i(i+1)}) = \frac{1}{\ln 10}\ln(1+\frac{q}{100i(i+1)}) < \frac{1}{\ln 10} \cdot (1+\frac{q}{100i(i+1)}) = \frac{q}{100 \ln 10}(\frac{1}{i-1} - \frac{1}{i})$.